

Bitcoins and Cryptocurrencies

Note: most of the slides used in this course are derived from those available for the book “Bitcoins and Cryptocurrencies Technologies – A Comprehensive Introduction”, Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller & Steven Goldfeder, 2016, Princeton University Press.

Lecture 5

Bitcoin mining

Recap: Bitcoin miners

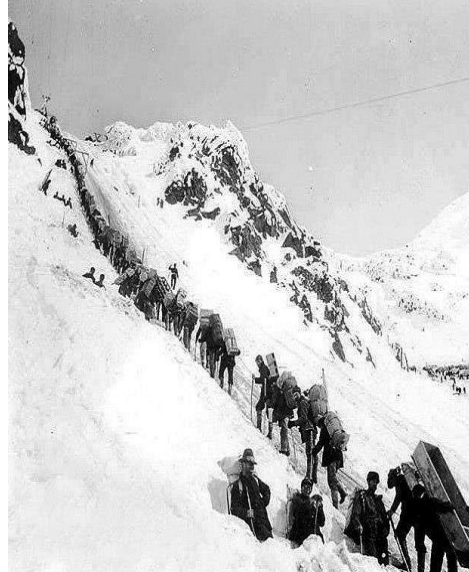
Bitcoin depends on miners to:

- Store and broadcast the block chain
- Validate new transactions
- Vote (by hash power) on consensus

But who are the miners? How did they get into this? How do they operate? What is there business model?...

The task of Bitcoin miners

So you want to be a miner?



- Some Gold miners of Klondike gold rush struck it rich, others lost everything
- Bitcoin mining has similar challenges as gold rush and other get-rich-quick schemes!

Klondike gold rush of 1898 - Gold miners ascending the Chilkoot pass

Mining Bitcoins in 6 easy steps

1. Join the network, listen for transactions

a. Validate all proposed transactions

2. Listen for new blocks, maintain block chain

a. When a new block is proposed, validate it

3. Assemble a new valid block

4. Find the nonce to make your block valid

5. Hope everybody accepts your new block

6. Profit!

Useful to
Bitcoin
network



```
graph LR; U[Useful to Bitcoin network] --> 1[1. Join the network, listen for transactions]; U --> 2[2. Listen for new blocks, maintain block chain]; U --> 3[3. Assemble a new valid block]; I[Incentivize miners to do above] --> 4[4. Find the nonce to make your block valid]; I --> 5[5. Hope everybody accepts your new block]; I --> 6[6. Profit!]
```

Incentivize
miners to
do above

Mining Bitcoins in 6 easy steps

1. Listen for transactions:

- a. Validate them by checking that signatures are correct
- b. Outputs being spent haven't been spent before.

2. Maintain block chain and listen for new blocks:

- a) Start by asking other nodes to give you all of the historical blocks that are already part of the block chain before you joined the network.
- b) Listen for new blocks that are being broadcast to the network.
- c) Must validate each block that you receive — by validating each transaction in the block and checking that the block contains a valid nonce.

3. Assemble a candidate block:

- a) Once you have an up-to-date copy of the block chain, you can begin building your own blocks.
- b) Group transactions that you heard about into a new block that extends the latest block you know about.
- c) Must make sure that each transaction included in your block is valid.

Mining Bitcoins in 6 easy steps

4. Find a nonce that makes your block valid:

- a) This step **requires the most work** and it's where all the **real difficulty happens for miners**.

5. Hope your block is accepted:

- a) Even if you find a block, there's **no guarantee that your** block will become part **of the consensus chain**.
- b) You have to hope that other miners accept your block and start mining on top of it, instead of some competitor's block

6. Profit:

- a) If all other miners do accept your block, then you profit!
- b) In early 2015, the **block reward is 25 bitcoins**.
- c) If any of the transactions in the **block contained transaction fees (about 1% of block rewards)**, the **miner collects those too**.

Finding a valid block

- Finding a nonce that makes your block valid.
- Two main hash-based structures
 - There's the block chain where each block header points to the previous block header in the chain.
 - Then within each block there's a Merkle tree of all of the transactions included in that block.
- The first thing that you do as a miner is to compile a set of valid transactions that you have from your pending transaction pool into a Merkle tree.
- In the block header, there's a 32 bit nonce field, and you keep trying different nonces looking for one that causes the block's hash to be under the target — roughly, to begin with the required number of zeros.

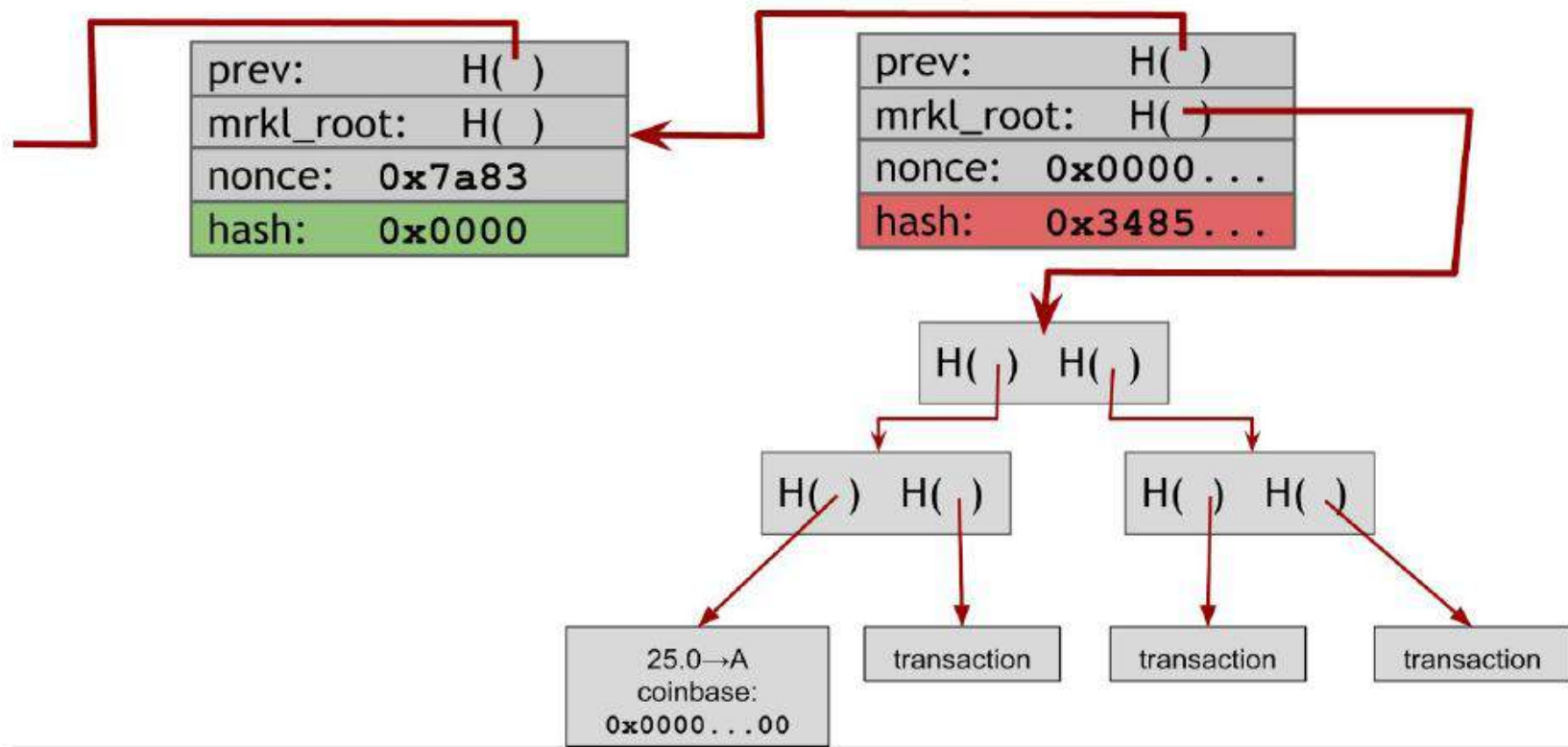


Figure 5.1: Finding a valid block. In this example, the miner tries a nonce of all 0s. It does not produce a valid hash output, so the miner would then proceed to try a different nonce.

Finding a valid block

- Changing the extra nonce in the coinbase transaction is much more expensive operation than changing the nonce in the block header
- For this reason, miners spend most of their time changing the nonce in the block header and only change the coinbase nonce when they have exhausted all of the 2^{32} possible nonces in the block header without finding a valid block.

Finding a valid block

prev:	H()
mrkl_root:	H()
nonce:	0x7a83
hash:	0x0000

prev:	H()
mrkl_root:	H()
nonce:	0xf77e...
hash:	0x0000...

All changed

H() H()

H() H()

H() H()

12.5→A
coinbase:
0x3df5...65

transaction

transaction

transaction

Mining difficulty

Difficulty. Exactly how difficult is it to find a valid block? As of March 2015, the mining difficulty target (in hexadecimal) is:

00000000000000000172EC000

so the hash of any valid block has to be below this value. In other words only one in about 2^{67} nonces that you try will work, which is a really huge number. One approximation is that it's greater than the human population of Earth squared. So, if every person on Earth was themselves their own planet Earth with seven billion people on it, the total number of people would be close to 2^{67} .

Is everyone solving the same puzzle?

- The answer is no! Firstly, it's unlikely that miners will be working on the exact same block as each miner will likely include a somewhat different set of transactions and in a different order.
- But more importantly, even if two different miners were working on a block with identical transactions, the blocks would still differ.
- Recall that in the coinbase transaction, miners specify their own address as the owner of the newly minted coins.
- This address by itself will cause changes which propagate up to the root of the Merkle tree, ensuring that no two miners are working on exactly the same puzzle unless they share a public key.
- This would only happen if the two miners are part of the same mining pool, in which case they'll communicate to ensure they include a distinct nonce in the coinbase transaction to avoid duplicating work.

Is everyone solving the same puzzle?

No!

Unlikely miners working on the same set of transactions (and thus the same block)

Even if some miners have the same transactions in the block, there is one which is not the same.

Q1: Do you know which one?

Answer: Coinbase!

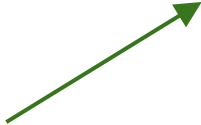
Q2: Why?

Answer: Miners will specify their own address in the output (hopefully each miner will use a different address!)

Setting the mining difficulty

Every two weeks (or approx. 2,016 blocks), compute:

```
next_difficulty= previous_difficulty *  
                  (2 weeks)/(time to mine last 2016 blocks)
```



Expected number of blocks in 2 weeks at 10 minutes/block

Setting the mining difficulty

- If the period were much shorter, the difficulty might fluctuate due to random variations in the number of blocks found in each period.
- If the period were much higher, the network's hash power might get too far out of balance with the difficulty.
- Each Bitcoin miner independently computes the difficulty and will only accept blocks that meet the difficulty that they computed.
- Miners who are on different branches might not compute the same difficulty value, but any two miners mining on top of the same block will agree on what the difficulty should be. This allows consensus to be reached.

Mining difficulty over time

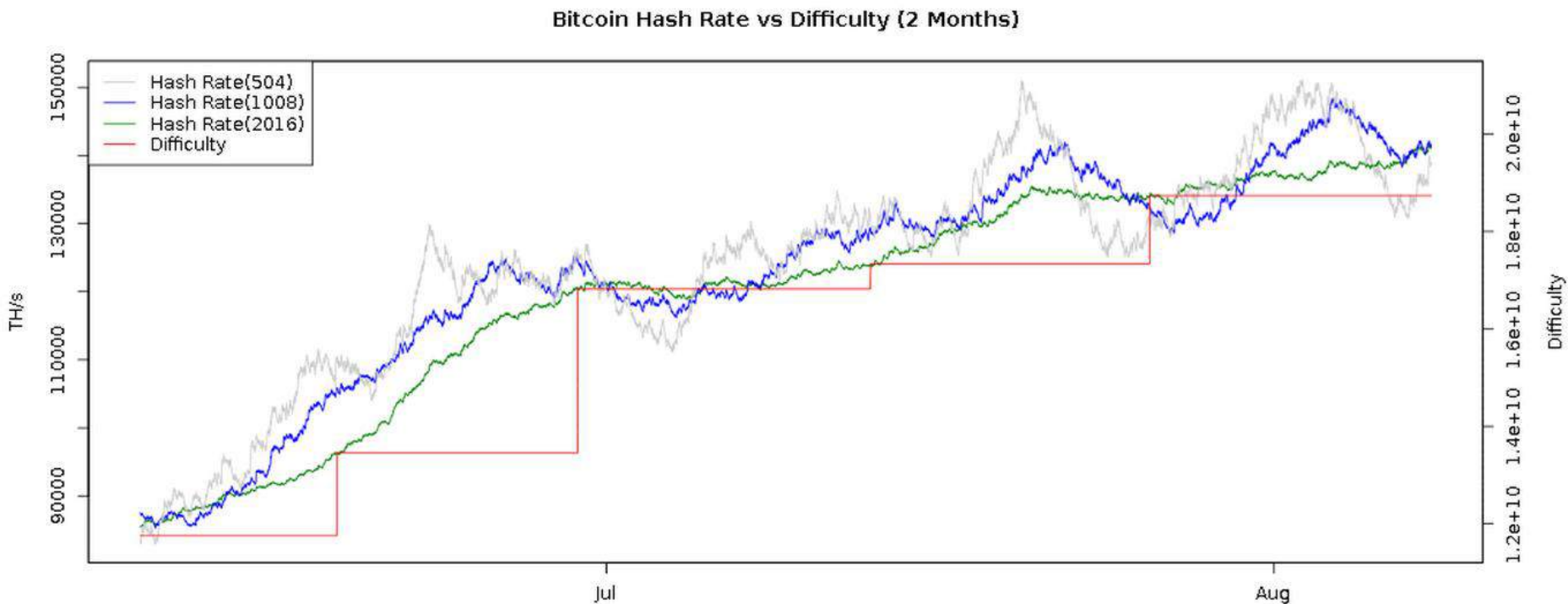


Figure 5.3: Mining difficulty over time (mid-2014). Note that the y-axis begins at 80,000 TH/s.

Mining difficulty over time

- Mining difficulty is affected by factors like how many new miners are joining, which in turn may be affected by the current exchange rate of Bitcoin.
- Generally, as more miners come online and mining hardware gets more efficient, blocks are found faster and the difficulty is increased so that it always takes about ten minutes to find a block.

Mining difficulty over time

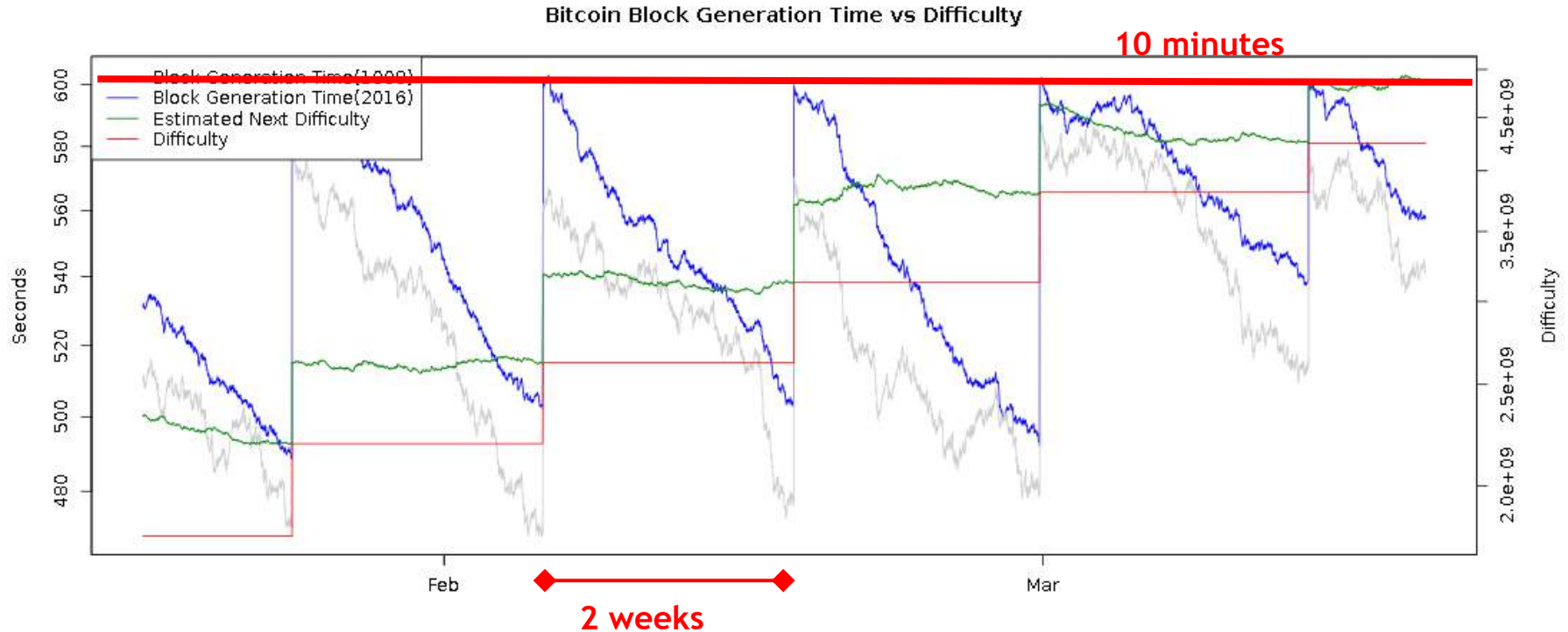
Factor(s) affecting mining difficulty:

- How many new miners join the network ← current exchange rate of Bitcoins
- Efficiency of mining hardware

***As more miners join and mining hardware becomes efficient → blocks are found faster → difficulty is increased!
Goal: keep the difficult at approx. 10 min per block***

Time to find a block

Figure 5.4 (a) : Time to find a block (early 2014). Note that the y-axis begins at 460 seconds. Due to continued rapid growth in mining power during this time, the time to find a block decreased steadily within each two-week window.



Bitcoin Block Generation Time vs Difficulty

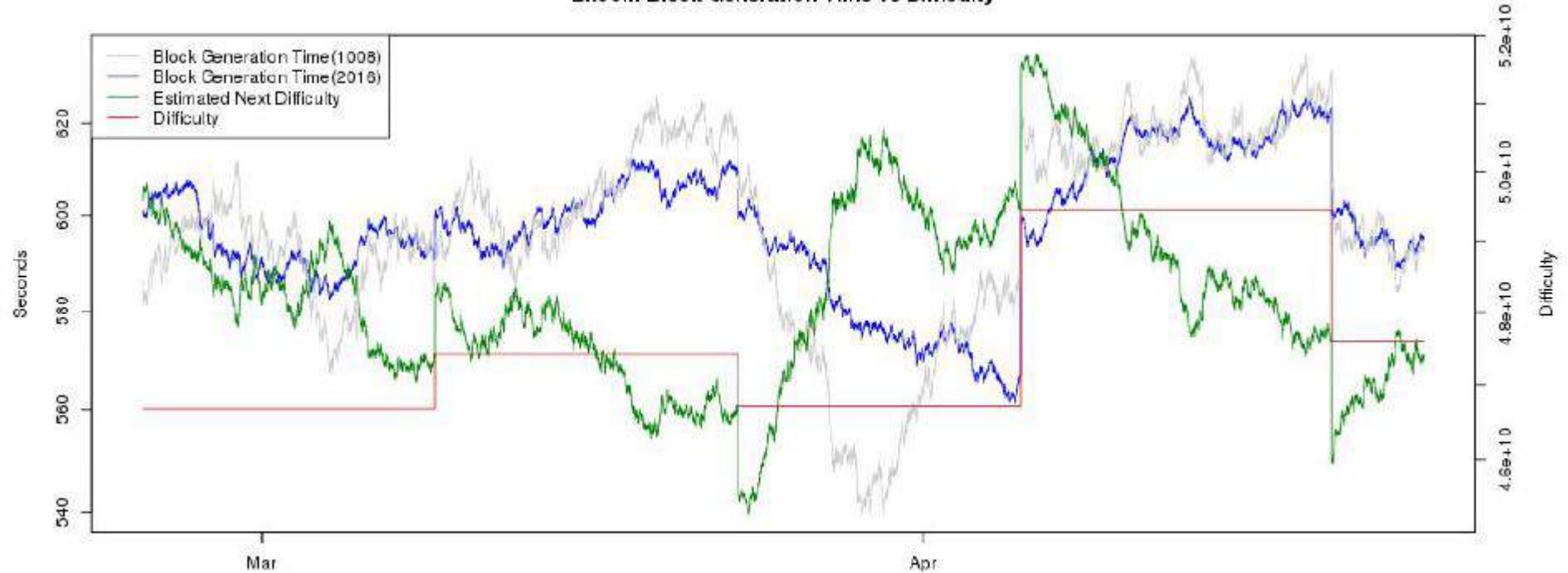


Figure 5.4 (b) : Time to find a block (early 2015). Note that the y-axis begins at 540 seconds. As the growth of the network has slowed, the time to find each block is much closer to 10 minutes and is occasionally over during periods where the network's hash power actually shrinks. Source: bitcoinwisdom.com

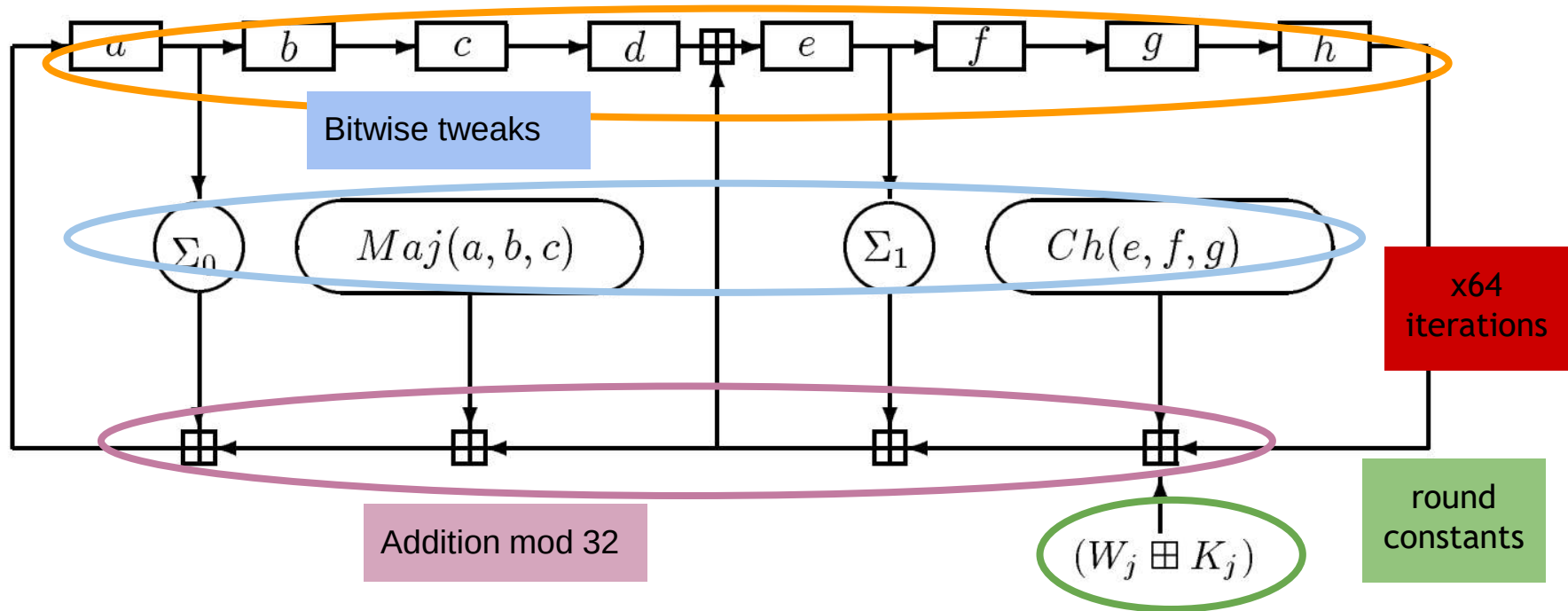
Mining hardware

SHA-256

- General purpose hash function
 - Part of SHA-2 family: SHA-224,SHA-384,SHA-512
- Published in 2001
- Designed by the NSA
- Remains unbroken cryptographically
 - Weaknesses known
- SHA-3 (replacement) under standardization

SHA-256 in more depth

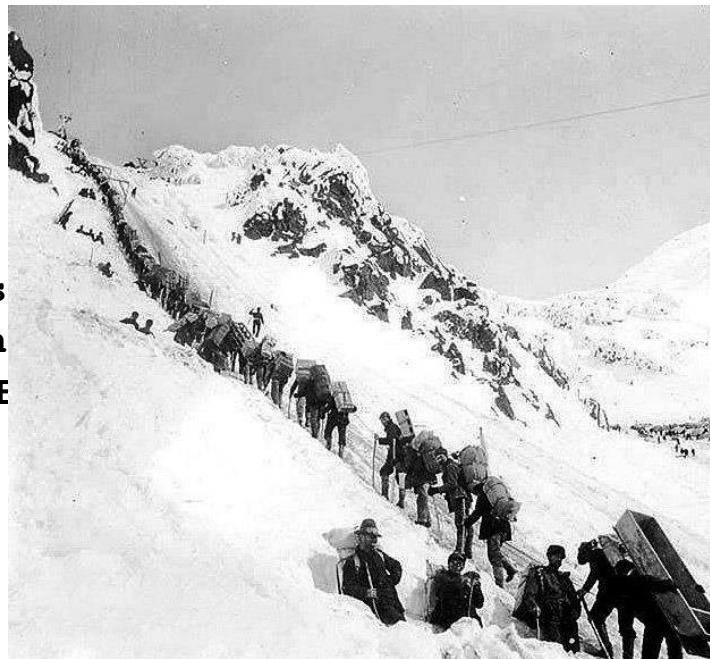
256-bit state



CPU mining

```
TARGET = (65535 << 208)/DIFFICULTY;  
coinbase_nonce = 0;  
while (1){  
    header = makeBlockHeader(trans  
    for (header_nonce=0;header_non  
        if (SHA256(SHA256(makeB  
TARGET)  
        break; //block  
    }  
    coinbase_nonce++;  
}
```

↑
two hashes



Throughput on a high-end PC $\approx$ **20 million hashes/sec**
 $\approx$ **139,461** years to find a block today!

GPU mining



- GPUs designed for high-performance graphics
 - High parallelism
 - High throughput
- First used for Bitcoin mining in October 2010
- Implemented in OpenCL
 - OpenCL: General purpose high level language for implementing non-graphic applications on GPUs

GPU mining advantages

- Easily available, easy to set up
- Parallel ALUs
 - Simultaneous SHA-256 computations
- Some GPUs support instructions for bit-wise operations
 - Useful for SHA-256 computations
- Can drive many from 1 CPU
- Can overclock!
 - Run them faster than they are actually designed for
 - Only downside is that overclocking might introduce errors (but errors can be quickly verified) or overheat
 - Introduction of some errors to gain a speed-up in hash computation is a good trade-off

“Goodput”

Observation: *some* errors are okay (may miss a valid block)

Goodput: throughput $\times$ success rate

Worth over-clocking by 50% with 30% errors!



Source:
LeonardH,
cryptocurrencie
stalk.com

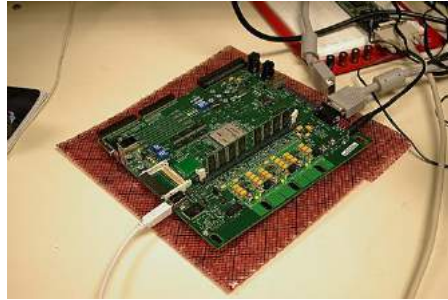
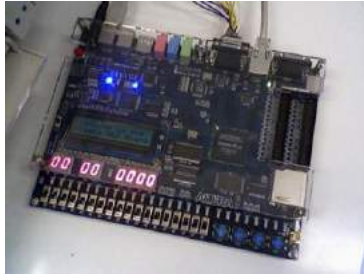
GPU mining disadvantages

- Poor utilization of hardware
 - Contains FPU's which are not used for SHA-256 at all
- Poor cooling
 - Cannot be stacked
- Large power draw
- Few boards to hold multiple GPUs
 - May need customized boards

Throughput on a good card = **20-200 MH/s** $\approx$ 200 million hashes/sec
 $\approx$ 173 years to find a block w/100 cards (at 2015 difficulty level)!

GPU mining is basically dead for Bitcoins ☹️

FPGA mining



- **F**ield **P**rogrammable **G**ate **A**rea
- First used for Bitcoin mining approx. June 2011
- Implemented in Verilog

FPGA mining advantages

- Higher performance than GPUs
 - Excellent performance on bitwise operations
- Better cooling
- Extensive customisation, optimisation



Bob Buskirk, thinkcomputers.org

FPGA mining disadvantages

- Higher power draw than GPUs designed for
 - Frequent malfunctions, errors
- Poor optimization of 32-bit adds
 - Critical for SHA-256 operations
- Less accessible
 - Cannot buy them at regular stores
 - Fewer hobbyists with sufficient expertise to program them
- More expensive than GPUs
- Marginal performance/cost advantage over GPUs

Throughput on a good card = 100-**1000 MHz** $\approx$ **1 billion hashes / sec**
25 years to find a block w/100 boards (at 2015 difficulty level)!

FPGA mining is also dead for Bitcoins – was even more short-lived than GPUs 😞

Bitcoin Application-specific Integrated Circuits (ASICs)

TerraMiner™ IV – 2TH/s Networked ASIC Miner

\$5,999

Shipping June 2014



300 GH Bitcoin Mining Card

The Monarch BPU 300 C

\$1,497.00

Qty:

1

ADD TO CART



Pre-Order Terms: This is a pre-order. 28nm ASIC bitcoin mining hardware products are shipped according to placement in the order queue, and delivery may take 3 months or more after order. All sales are final.



DETAILS :

- 2,5 TH/s
- Dimensions:
15" x 13.3" x 13.7"
(38cm x 34cm x 35cm)
- 28nm ASIC technology
- Silent Cooling
- In-built WiFi Connection
(without Antenna)
- Less than 750 watt (0.3 per GH)
- 1 Year Guarantee
- \$ 5.800

COMES WITH :

1. Power Supply
2. Free Remote Power Outlet & Smartphone App
3. Free User Guide
4. Free Personal Assistance for Setup

SHIPPING :

- Worldwide, Express
- Included in the price
- Available:
100 Units; Shipping April
(Week 3)

Bitcoin ASICs

- Special purpose
 - Approaching known limits on feature sizes
 - Less than 10x performance improvement expected
- Designed to be run constantly for life
- Require significant expertise, long lead-times
- Perhaps the fastest chip development ever!

Case study: TerraMiner IV



- First shipped Jan 2014
- 2 TH/s
- Cost: US\$6,000

Still, 14 months to find a block!

Market dynamics (2013/2014)

- Most boards obsolete within 3-6 months
 - Half of profits made in first 6 weeks
- Shipping delays are devastating to customers
- Most companies require pre-orders
- Most individual customers should have lost...

But... rising prices have saved them!

Professional mining centers

Needs:

- Cheap power
- Good network
- Cool climate
 - (Georgia and Iceland are popular destinations for bitcoin mining)



BitFury mining center, Republic of Georgia

Evolution of mining



CPU



GPU



FPGA



ASIC



Gold pan



Sluice box



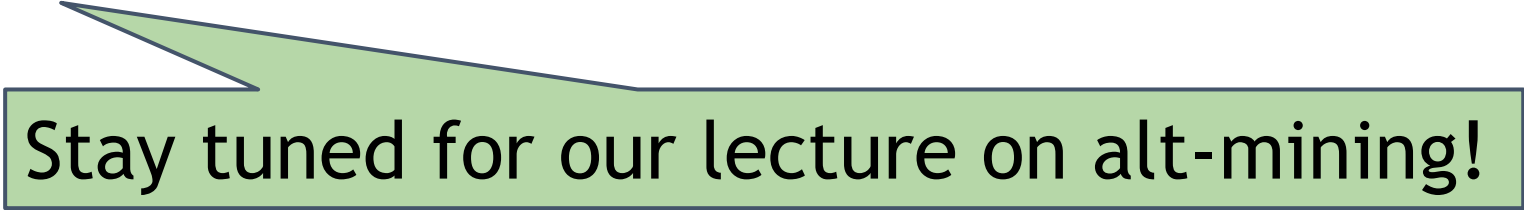
Placer mining



Pit mining

The future

- Can small miners stay in the game?
- Do ASICs violate the original Bitcoin vision?
- Would we be better off without ASICs?



Stay tuned for our lecture on alt-mining!

Energy consumption & ecology

Thermodynamic limits

Landauer's principle: Any non-reversible computation must consume a minimum amount of energy.

Specifically, each bit changed requires $(kT \ln 2)$ joules

SHA-256 is not reversible

Energy consumption is inevitable

Energy aspects of Bitcoin mining

- **Embodied energy:** used to manufacture mining chips & other equipment
 - Should decrease over time
 - Returns to scale
- **Electricity:** used to perform computation
 - Should increase over time
 - Returns to scale
- **Cooling:** required to protect equipment
 - Costs more with increased scale!

Estimating energy usage: top-down

- Each block worth approximately **US\$6,500**
- Approximately **\$11/s** generated
- Industrial electricity (US): **\$0.03/MJ**
 - **Equivalently, \$0.10/kWh**

If Bitcoin miners were spending all \$11 per second of earnings on buying electricity

Upper bound on electricity consumed:

$$367 \text{ MJ/s} = 367 \text{ MW}$$

Estimating energy usage: bottom-up

- Best claimed efficiency:
 - Most cutting edge ASIC - 3 gigahashes/sec per watt (or 3 billion hashes per second for 1 watt of power consumed)
- Network hash rate: **350,000,000** gigahashes/sec
 - Excludes cooling and embodied energy

Lower bound on electricity consumed (to produce that many hashes per second at that efficiency):

150 MW

How much is a MW?



Three Gorges Dam = 10,000 MW
typical hydro plant $\approx$ 1,000 MW

Kashiwazaki-Kariwa
nuclear power plant = 7,000 MW
typical nuclear plant $\approx$ 4,000 MW



major coal-fired plant $\approx$ 2,000 MW

In summary – Bitcoin energy usage

- The entire Bitcoin network consumes roughly 10% of a large power plant's output
- Significant but still small compared to other usages of electricity

Is Bitcoin mining wasteful?

- All payment systems require energy!



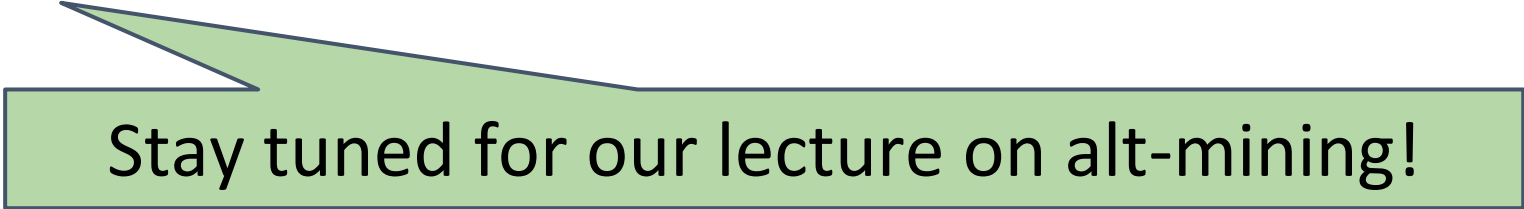
- Although, it may be nice to have a currency with a less energy-intensive puzzle, but the same level of security!

Data furnaces

- **Observation:** in the limit, computing devices produce heat almost as well as electric heaters!
- Why not install mining rigs as home heaters?
- Challenges:
 - Ownership/maintenance model
 - Gas heaters still at least 10x more efficient
 - What happens in summer?

Open questions

- Will Bitcoin drive out electricity subsidies?
- Will Bitcoin require guarding power outlets?
- Can we make a currency with no proof-of-work?



Stay tuned for our lecture on alt-mining!

Mining pools

Economics of being a small miner



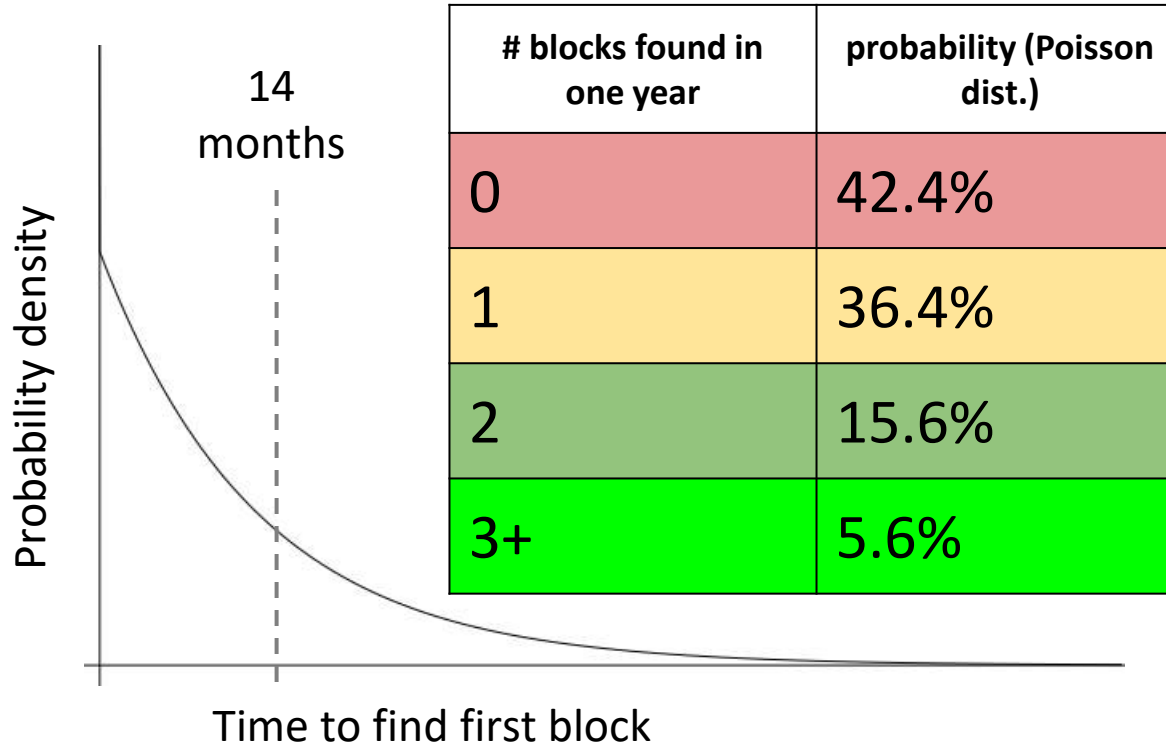
TerraMiner IV

- Cost of a mining rig: $\approx$ US\$6,000
- Expected time to find a block: $\approx$ 14 months
- In 2015, a block was worth $\approx$ \$10,000
- Amortized expected revenue: $\approx$ \$400/month (after factoring in electricity and other operating costs)

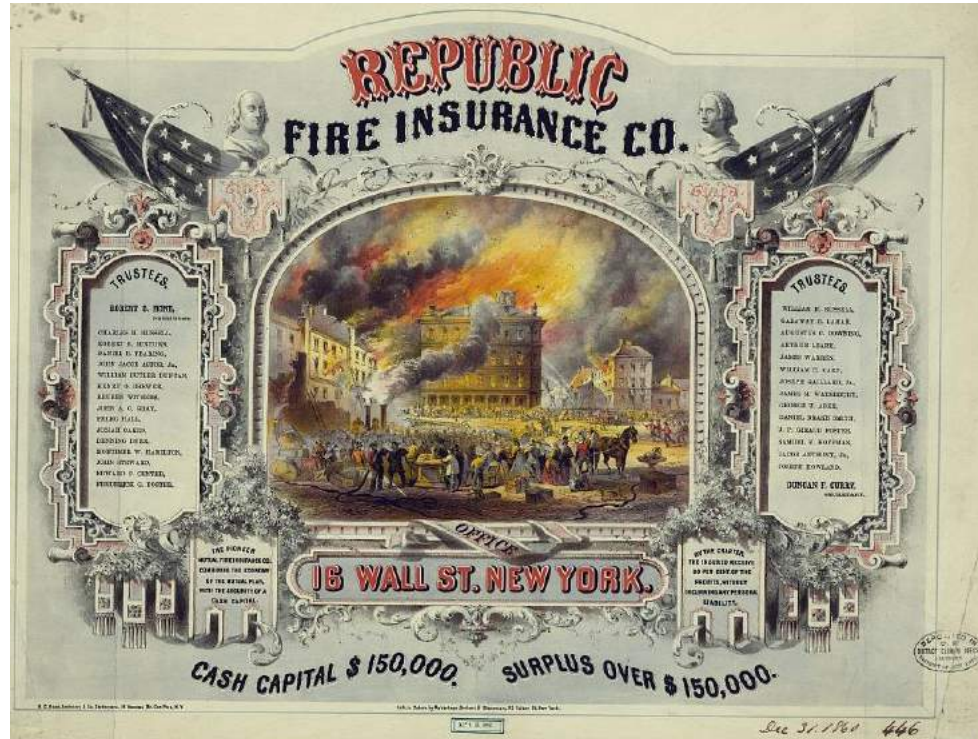
Given the above numbers, it may make sense to buy the mining rig, but remember mining is a random process $\rightarrow$ possible miner may never be able to mine a block $\rightarrow$ make any money

Figure 5.11: Illustration of uncertainty in mining. Assuming that the global hash rate is constant and the mean time to find a block is 14 months, the variance for a small miner is quite high.

Mining uncertainty



Idea: could small miners pool risk?



Mining pools

- Historically, when small business people faced a lot of risk, they formed mutual insurance companies to lower that risk.
- Farmers, for example, would get together and agree that if any individual farmer's barn burned down the others would share their profits with that farmer.
- Could we have a mutual insurance model that works for small Bitcoin miners?

Mining pools

- A mining pool is exactly that — mutual insurance for Bitcoin miners.
- A group of miners will form a pool and all attempt to mine a block with a designated coinbase recipient. That recipient is called the pool manager.
- So, no matter who actually finds the block, the pool manager will receive the rewards.
- The pool manager will take that revenue and distribute it to all the participants in the pool based on how much work each participant actually performed.
- Of course, the pool manager will also probably take some kind of cut for their service of managing the pool.

Mining pools

- **Goal:** pool participants all attempt to mine a block with the same coinbase recipient (aka pool manager)
 - Send money to key owned by pool manager
- Distribute revenues to members based on how much work they have performed
 - Minus a cut for pool manager

How do we know how much work members perform?

Mining pools

- Assuming everybody trusts the pool manager, **this works great for lowering miners' variance.**
- But how does a pool manager know how much work each member of the pool is actually performing?
- How can the pool manager divide the revenue commensurate with the amount of work each miner is doing?
- Obviously the pool manager doesn't want to just take everyone's word for it because people might claim that they've done more than they actually did.

Mining shares

- There's an elegant solution to this problem. **Miners can prove probabilistically how much work they're doing by outputting shares , or near-valid blocks.**
- Say the **target is a number beginning with 67 zeros.**
- A **block's hash must be lower than the target for the block to be valid.**
- In the process of searching for such a block, miners will find some blocks with **hashes beginning with a lot of zeros, but not quite 67.**
- Miners can show these **nearly valid blocks** to prove that they are **indeed working.**
- A **share might require say 40 or 50 zeros,** depending on the **type of miners the pool is geared for.**

Mining shares

In this figure, the **dull green hashes are shares**, while the **bright green hash is from a valid block** (which is also a valid share).

Idea: prove work with “near-valid blocks” (shares)

```
4AA087F0A52ED2093FA816E53B9B6317F9B8C1227A61F9481AFED67301F2E3FB
D3E51477DCAB108750A5BC9093F6510759CC880BB171A5B77FB4A34ACA27DEDD
00000000008534FF68B98935D090DF5669E3403BD16F1CDFD41CF17D6B474255
BB34ECA3DBB52EFF4B104EBBC0974841EF2F3A59EBBC4474A12F9F595EB81F4B
00000000002F891C1E232F687E41515637F7699EA0F462C2564233FE082BB0AF
0090488133779E7E98177AF1C765CF02D01AB4848DF555533B6C4CFCA201CBA1
460BEFA43B7083E502D36D9D08D64AFB99A100B3B80D4EA4F7B38E18174A0BFB
000000000000000078FB7E1F7E2E4854B8BC71412197EB1448911FA77BAE808A
652F374601D149AC47E01E7776138456181FA4F9D0EEDD8C4FDE3BEF6B1B7ECE
785526402143A291CFD60DA09CC80DD066BC723FD5FD20F9B50D614313529AF3
000000000041EE593434686000AF77F54CDE839A6CE30957B14EDEC10B15C9E5
9C20B06B01A0136F192BD48E0F372A4B9E6BA6ABC36F02FCED22FD9780026A8F
```

Mining shares

- The pool manager will **also run a Bitcoin node on behalf of participants,** collecting transactions and assemble them into a block.
- The manager will include **their own address in the coinbase transaction and send the block to all of the participants in the pool.**
- All pool participants work on this block, and they prove **that they've been working on it by sending in shares.**

Mining shares

- When **a member of the pool finds a valid block**, they **sends it to the pool manager** who distributes the reward in proportion to the amount of work done.
- The miner who actually **finds the block is not awarded a special bonus**, so if **another miner did more work than**, that other miner will be paid more even though they weren't the one who ended up finding a valid block. See Figure 5.13.

Figure 5.13: Mining rewards. Three participants pictured here are all working on the same block. They are awarded commensurate with the amount of work done. Even though the miner on the right was the one to find the valid block, the miner on the left is paid more since this miner did more work. There is (typically) no bonus paid to the miner who actually finds the block.

Mining pools

Hey folks! Here's our next block to work on

Pool manager

prev:	H()
mrkl_root:	H()
nonce:	
hash:	

coinbase:
25→pool

0x000000000000a877902e...
\$\$\$

0x0000000000001e8709ce...

0x000000000000490c6b00...

0x00000000000007313f89...

0x00000000000000003f89...
\$

0x00000000000045a1611f...



Mining shares

- There are a few options for exactly how exactly the pool manager calculates how much to pay each miner based on the shares they submit.
- We'll look at two of the common, simpler ones.
- There are many other schemes that are also used, but these will illustrate the trade-offs between reward schemes.

Mining pool variations

- **Pay per share:** flat reward per share
 - Typically minus a significant fee
 - What if miners never send in valid blocks?
- **Proportional:** typically since last block
 - Lower risk for pool manager
 - More work to verify
- **“Luke-jr” approach:** no management fee

Pool Hopping

- Miners may be incentivized to switch pools (of different types) at different times.
- Consider that a purely proportional pool will **effectively pay out a larger amount per share if a block is found quickly**, as it always pays **one block reward** no matter **how long it has been since the last block was found**
- For instance, a clever miner might **try mining in a proportional pool early in the cycle** (while rewards per share are relatively high), **only to “hop” to a pay-per-share pool later** (while rewards per share are relatively low in the proportional pool).
- As a result of this, **proportional pools aren't really practical**.
- More complicated schemes, such as “**pay per last N shares submitted**” are more common, but even these are **subject to subtle pool hopping behavior**.
- **Open problem: How to design mining pool reward that is not vulnerable to manipulation!**

Mining pool protocols

- Mining pools first started around 2010 in the GPU era of Bitcoin mining.
- They instantly became very popular for the obvious reason that they lowered the variance for the participating miners.
- They've become quite advanced now. There are many protocols for how to run mining pools and it has even been suggested that these mining pool protocols should be standardized as part of Bitcoin itself.

Mining pool protocols

- Just like there's a Bitcoin protocol for running the peer-to-peer network, mining pool protocols provide a **communication API for the pool manager to send all of the members the details of the block to work on** and for **the miners to send back to the pool manager the shares that they're finding**.
- **getblocktemplate (GBT)** is officially standardised as a Bitcoin Improvement Proposal (BIP).
- A competing protocol, **Stratum**, is currently **more popular in practice** and is a proposed BIP.
- Unlike the Bitcoin protocol itself, it is only a **minor inconvenience to have multiple incompatible mining pool protocols**.
- Each pool can simply pick whichever protocol they like and the market can decide.

Mining pool protocols

- Some **mining hardware even supports these protocols** at the hardware level, which will ultimately **limit their development flexibility somewhat**.
- However, this makes it very simple to **buy a piece of mining hardware and join a pool**.
- You just plug it into the wall — both the electricity and your network connection — choose a pool, and then it will **start immediately getting instructions from the pool, mining and converting your electricity into money**.

Mining pool protocols

- API for fetching blocks, submitting shares
 - Stratum
 - Getwork
 - Getblockshare
- Proposed for standardization with a BIP (Bitcoin Improvement Proposal)
- Increasingly important; some hardware support

51% mining pools

- As of early 2015, the vast majority of all miners are mining through pools with very few miners mining “solo” anymore.
- In June 2014, GHash.io, the largest mining pool, got so big that it actually had over 50% of the entire capacity over the Bitcoin network.
- Essentially GHash offered such a good deal to participating miners that the majority wanted to join.

51% mining pools

- This is something that people had feared for a long time and this led to a backlash against GHash.
- By August 2014, GHash's market share had gone down by design as they stopped accepting new participants.
- Still, two mining pools controlled about half of the power in the network

Mining pool history

- First pools appear in late-2010
 - Back in the GPU era!
- By 2014: around 90% of mining pool-based
- June 2014: GHash.io exceeds 50%

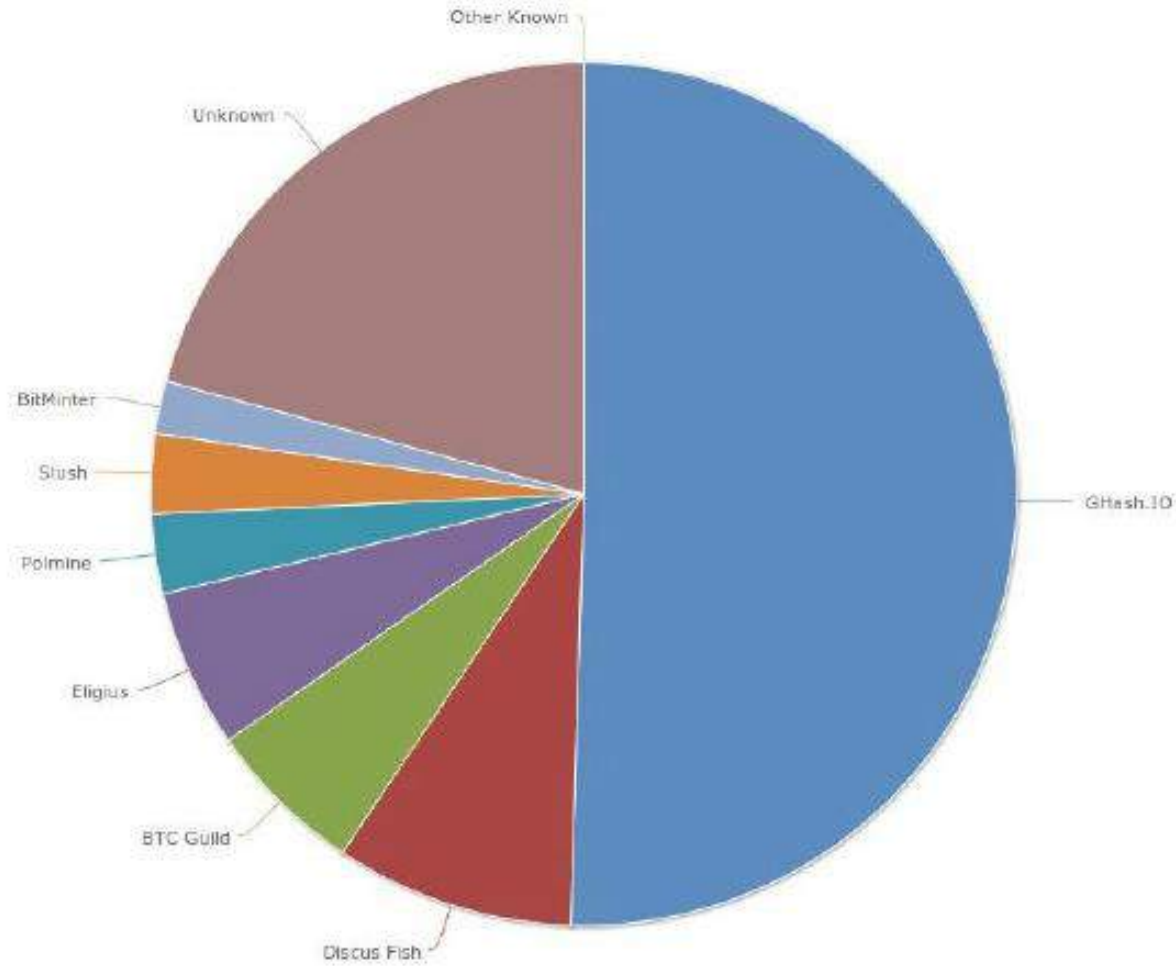


Figure 5.14 (a) Hash power by mining pool, via blockchain.info (June 2014)

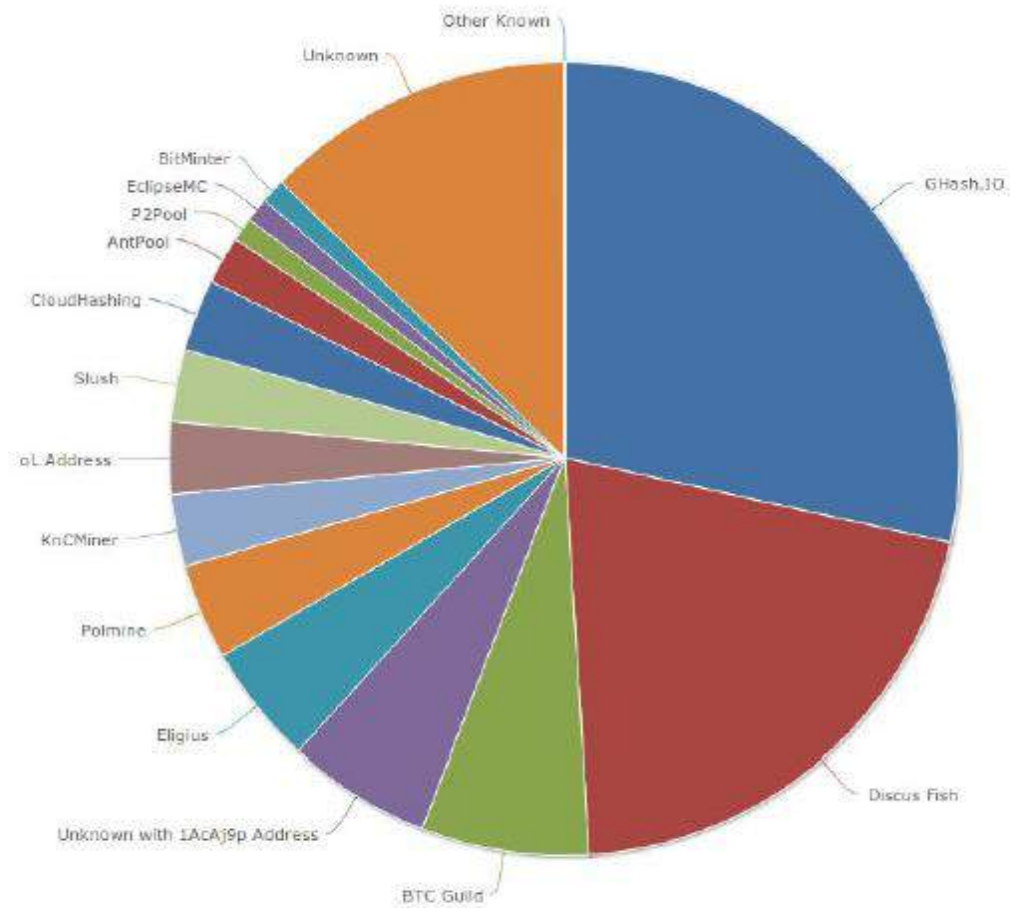


Figure 5.14 (b) Hash power by mining pool, via blockchain.info (August 2014)

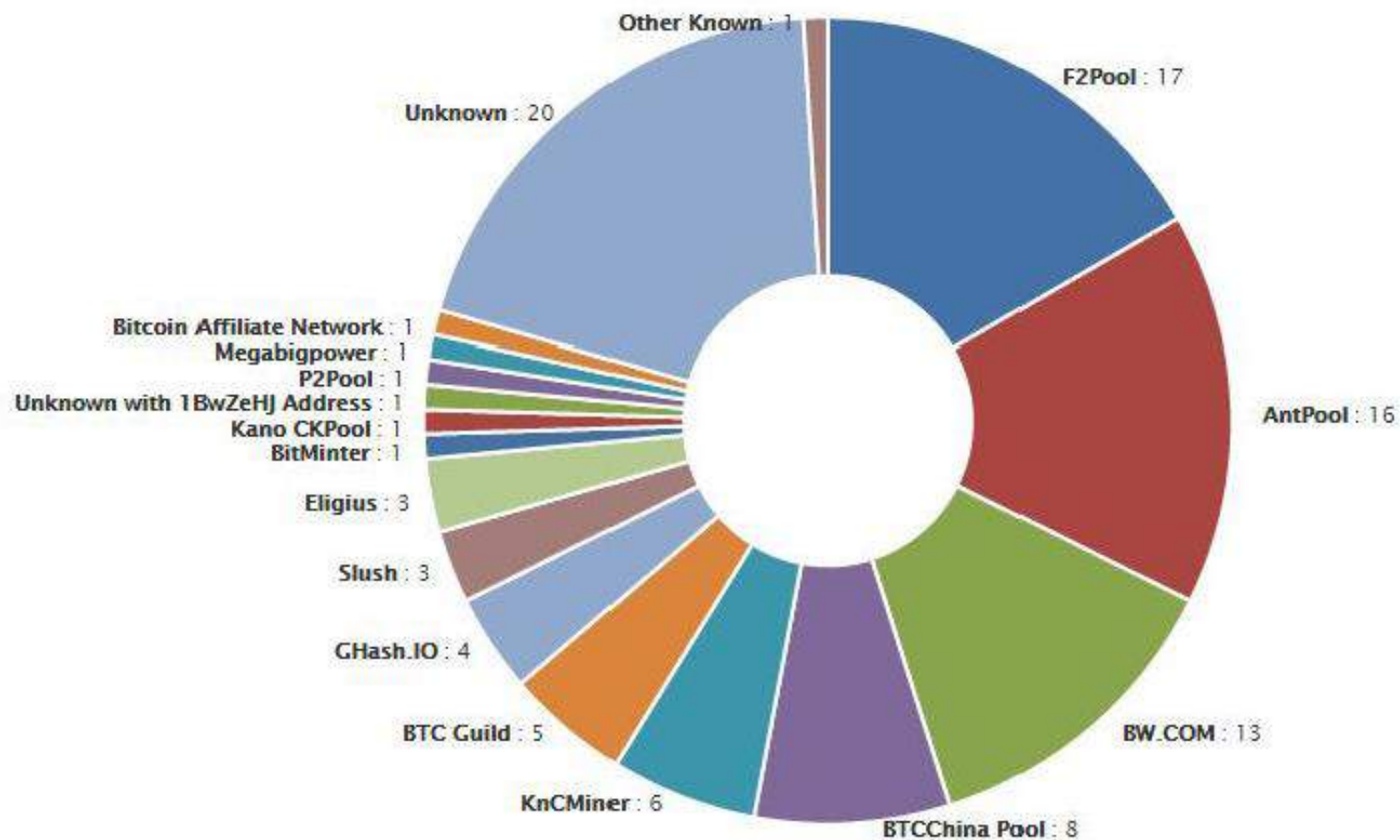


Figure 5.14 (c) Hash power by mining pool, via blockchain.info (April 2015)

51% mining pools

- It is worth noting that mining pools might be hiding actual concentration of mining power in the hands of **a few large mining organizations which can participate in multiple mining pools simultaneously** to hide their true size. **This practice is called laundering hashes.**
- It remains unknown how **concentrated physical control of mining hardware actually** is and **mining pools make this quite difficult to determine** from the outside.

Are mining pools a good thing?

- Pros
 - Make mining more predictable
 - Allow small miners to participate
 - More miners using updated validation software
- Cons
 - Lead to centralization (It's an open question how much power the operators of a large mining pool actually have.)
 - Discourage miners from running full nodes

Can we prevent pools?

Mining incentives and strategies

Game-theoretic analysis of mining

Several strategic decisions

- Which transactions to include in a block
 - Default: any above minimum transaction fee
- Which block to mine on top of
 - Default: longest valid chain
- How to choose between colliding blocks
 - Default: first block heard
- When to announce new blocks
 - Default: immediately after finding them

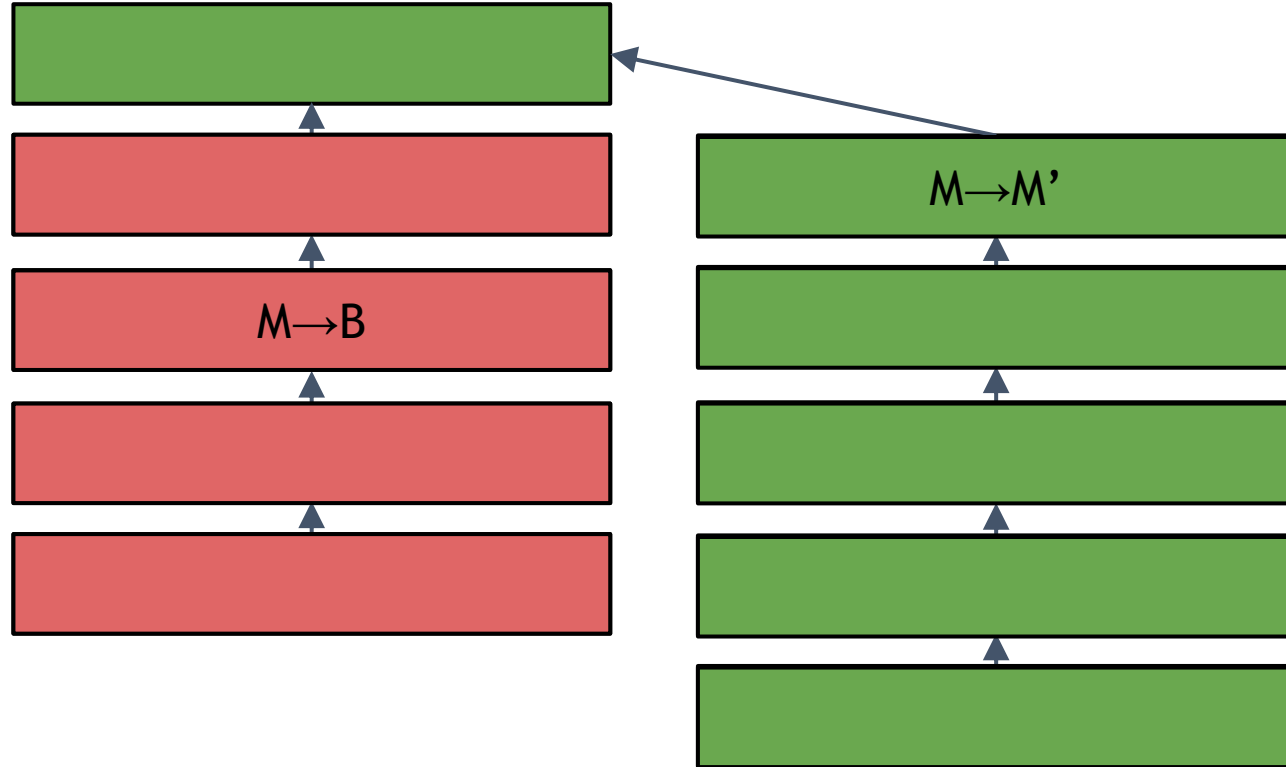
Game-theoretic analysis of mining

Assume you control $0 < \alpha < 1$ of mining power

Can you profit from a non-default strategy?

For some α , **YES**, though analysis is ongoing!

Forking attacks



Forking attacks

- Certainly possible if $\alpha > 0.5$
 - may be possible with less
 - avoid block collisions
- Attack is detectable
- Might be reversed
- Might crash exchange rate



Goldfinger Attack?

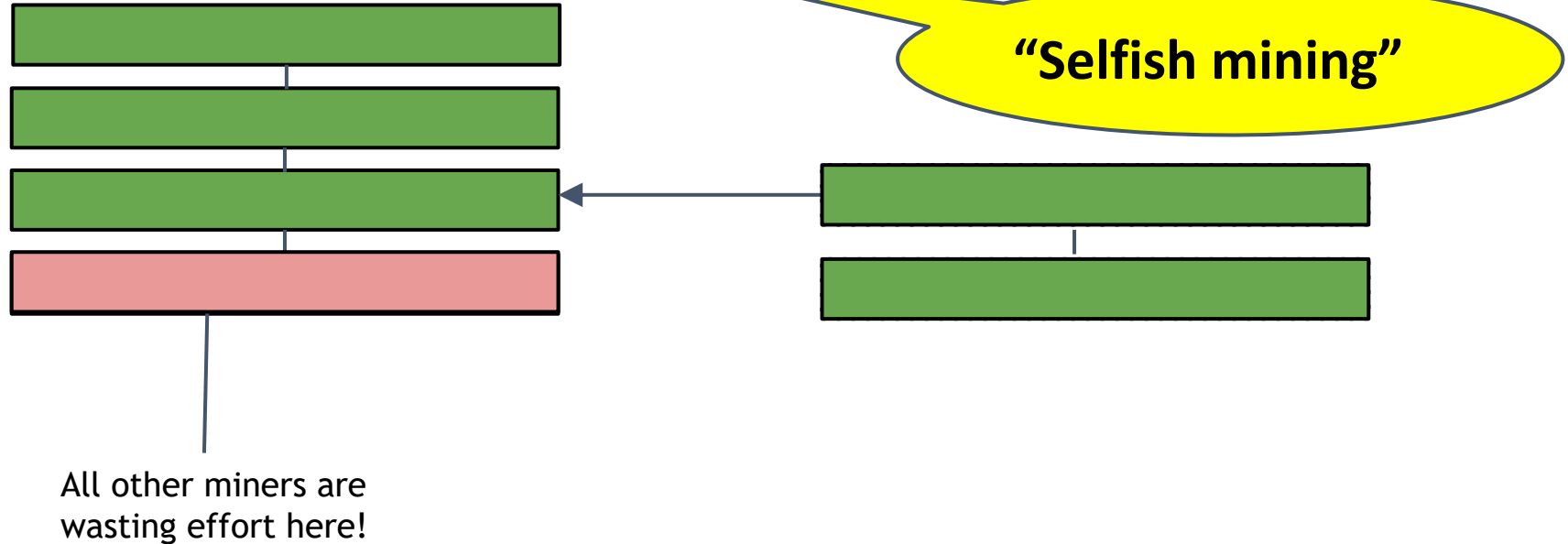
Forking attacks via bribery

- **Idea:** building $\alpha > 0.5$ is expensive. Why not rent it instead?
- Payment techniques:
 - Out-of-band bribery
 - Run a mining pool at a loss
 - Insert large “tips” in the block chain

Will miners accept bribes?
It is possible → Tragedy of Commons
This is an open problem!

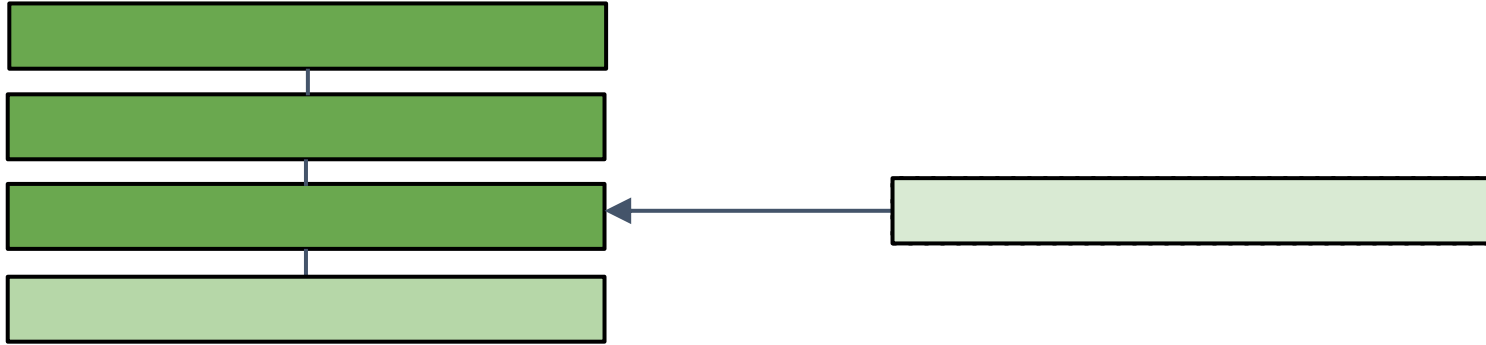
Block-withholding attacks

Strategy: don't announce blocks right away. Try to get ahead!



Block-withholding attacks, take 2

What happens if a block is announced when you're ahead by 1?



The race is on!

Block-withholding attacks

- Improved strategy for any α if you can win every race
 - Ideal network position
 - Bribery?
- With a 50% chance of winning races, improved strategy for $\alpha > 0.25$
- Not yet observed in practice!

Surprising departure from previous assumptions

Punitive forking

- Suppose you want to blacklist transactions from address X
 - Freeze an individual's money forever
- **Extreme strategy:** announce that you will refuse to mine on any chain with a transaction from X

With $\alpha < 0.5$, you'll soon fall behind the network

Feather-forking strategy

- **To blacklist transactions from X**, announce that you will refuse to mine *directly* on any block with a transaction from X
 - But you'll concede after n confirming blocks
- Chance of pruning an offending block is α^2

Response to feather forking

- **For other miners**, including a transaction from X induces an α^2 chance of losing a block
- Might be safer to join in on the blacklist
- Can enforce a blacklist with $\alpha < 0.5$!

Success depends on convincing other miners you'll fork

Feather-forking: what is it good for?

- Freezing individual bitcoin owners
 - ransom/extortion
 - law enforcement?
- Enforcing a minimum transaction fee...

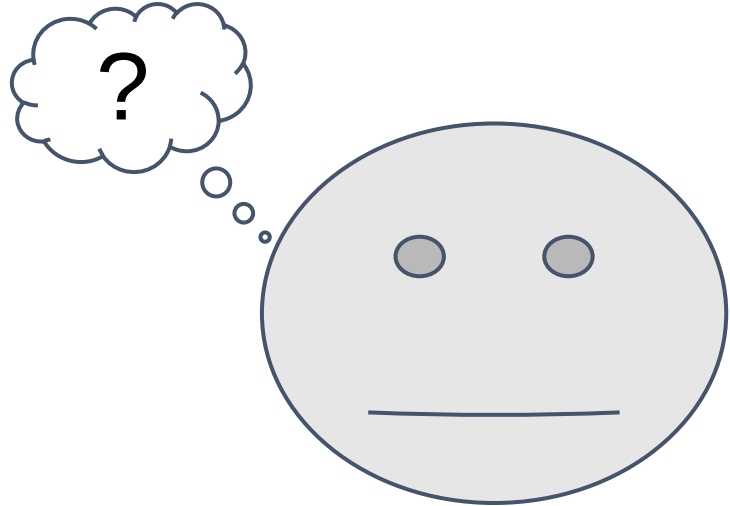
A second look at transaction fees

Default policy:

```
priority = sum(input_value * input_age) / size_in_bytes
```

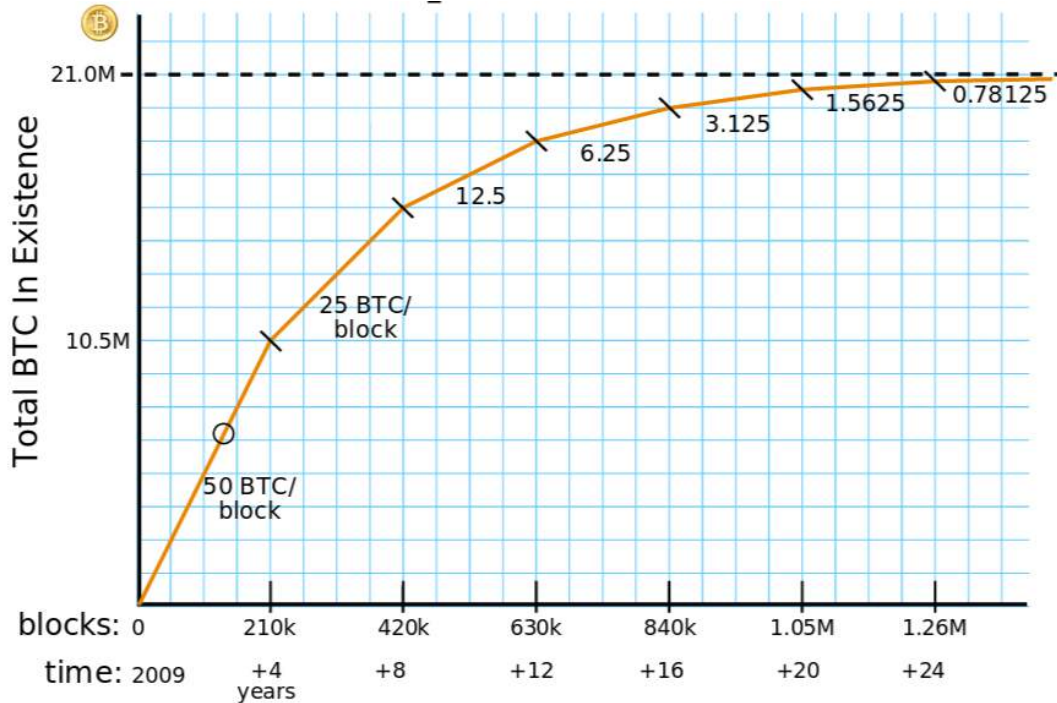
Accept without fees if:

```
priority > 0.576
```



Transaction fees will matter more

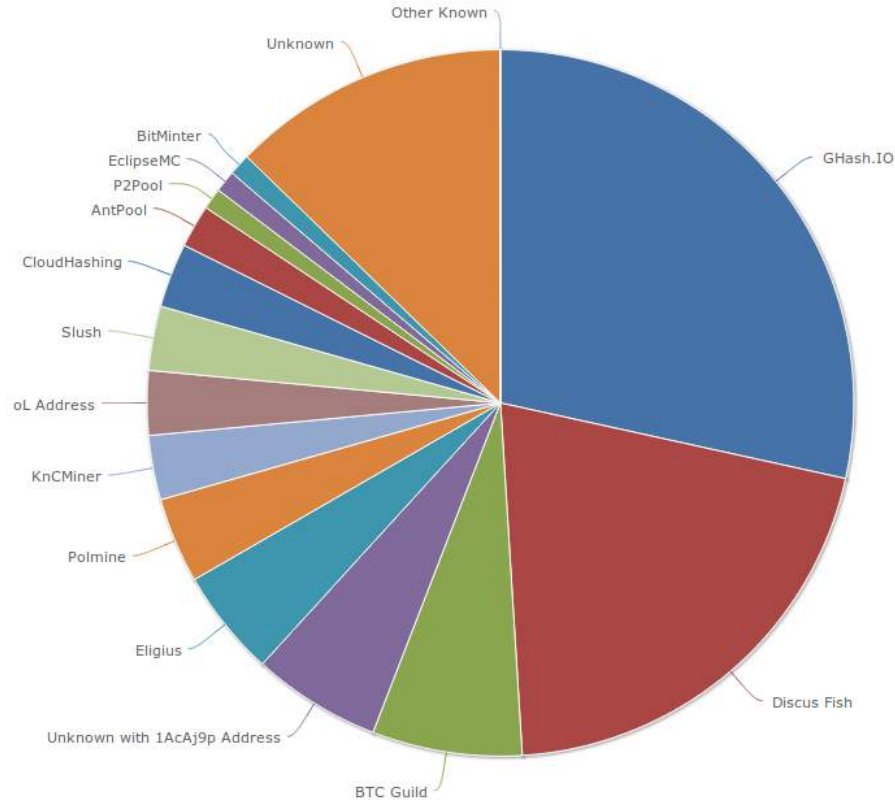
Currently, block rewards are > 99% of miner revenue. But:



**Eventually,
transaction fees
will dominate**

Courtesy: Brian Warner

Will miners cooperate to enforce fees?



Summary

- Miners are free to implement any strategy
- Very little non-default behavior in the wild
- No complete game-theoretic model exists

Things might be about to get interesting...

In the next lecture...

How much anonymity does
Bitcoin provide?